

Query q : perturb sub-plan cardinalities by estimation error σ

which regime? (size of the error)

Small error

condition number κ
distance to nearest
plan-switch wall

predicts regret
 $\rho \approx 0.70$ (beats q -error)

Large error

ACS_∞ (this work)
average over cost cells

predicts *which queries*
 $\rho \approx 0.54$ (q -
error: 0.05)

Worst case

MSO (Haritsa)
maximum over cells

worst-case bound
(prior art)

One cost-ratio spectrum $\{r_k\}$ of the candidate plans, three weightings:

$\kappa \sim$ nearest gap • $\text{ACS}_\infty = \sum_k r_k \pi_k$
(minimax-rank average) • $\text{MSO} = \max_k r_k$